Project 11:

Facial Recognition System

Problem Statement

Security systems relying on passwords, ID cards, or PINs are vulnerable to theft, forgery, and misuse. The ML Team is tasked with building a facial recognition system that can authenticate or identify individuals reliably, even under variations in lighting, pose, or facial expression.

Project Idea & Scope

Develop a face recognition model that can distinguish between individuals with high accuracy and low false acceptance. The system should be deployable in a simple application for testing real-time identification or authentication.

The work will include:

1. **Data Preparation**
    - Find a publicly available facial recognition dataset (e.g., LFW, VGGFace2).
    - Preprocess images: resize, normalize, and ensure consistent face cropping.
2. **EDA + Feature Building**
    - Explore dataset distribution across identities.
    - Apply data augmentation (flipping, rotation, scaling) to improve robustness.
3. **Model Training & Validation**
    - Train a CNN-based model or fine-tune a pre-trained model (e.g., FaceNet, VGG-Face).
    - Evaluate using accuracy, precision, recall, and False Acceptance Rate (FAR).
4. **Deployment via Web Interface**
    - Deploy the trained model in a simple web app (Streamlit or JS).
    - Allow users to upload or capture an image and get an identity verification result.

Deliverable

At the end of the project, the ML Team should deliver:

- A trained and validated facial recognition model with clear evaluation metrics.
- A short summary report documenting the approach and results.
- A demo application that performs real-time face recognition for testing.